College Physics 2

Electric Field

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Electric Field Principles

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OBJECTIVE

- State general properties of electric field.
- Interpret electric field using electric field lines.

Electric Field

Electric field is defined as the area where electrical force can be experienced.

A charged body creates an electric field in the space around it.

Electric Field

Suppose we have two charges A and B. B has a charge q_0 and experiences a force F_0 due to A. A, on the other hand, also experiences a force F_0 in the opposite direction.

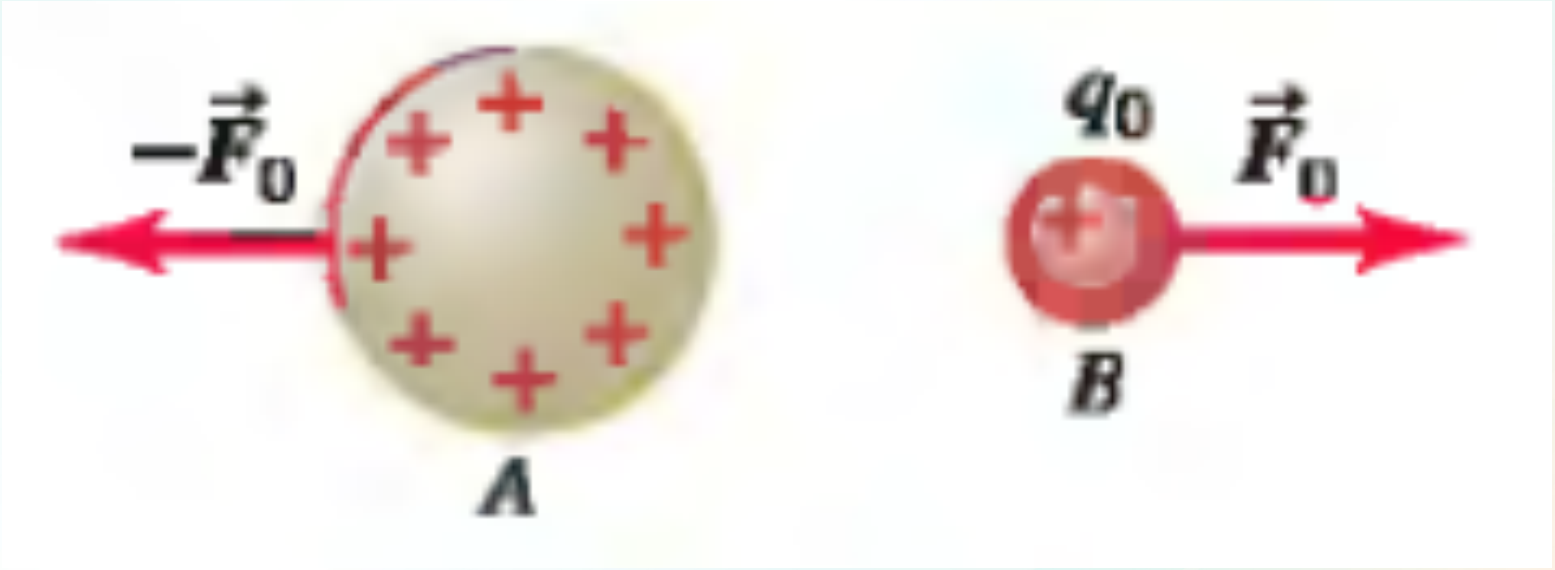


Figure 1. a) A charged body creates an electric field in the space around it. . From. *Sear's & Zemansky's University Physics With Modern Physics* (pp 699) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson.

Electric Field

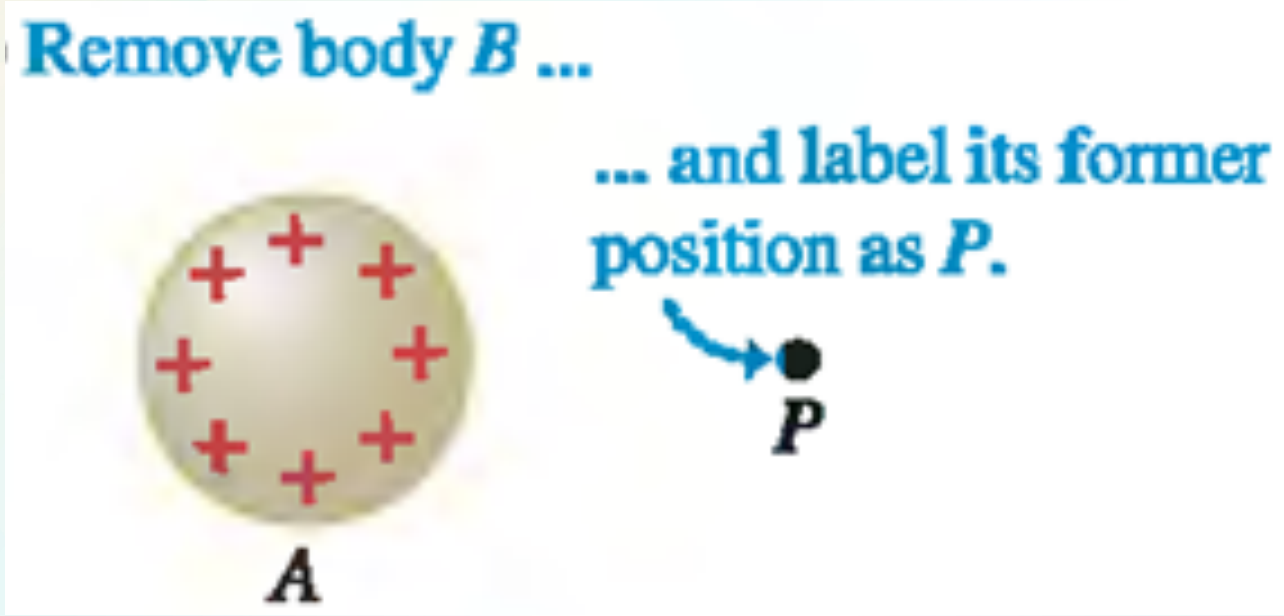


Figure 1. b) A charged body creates an electric field in the space around it. . From. *Sear's & Zemansky's University Physics With Modern Physics* (pp 699) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson.

Charge A modifies the properties of space around it.

Electric Field

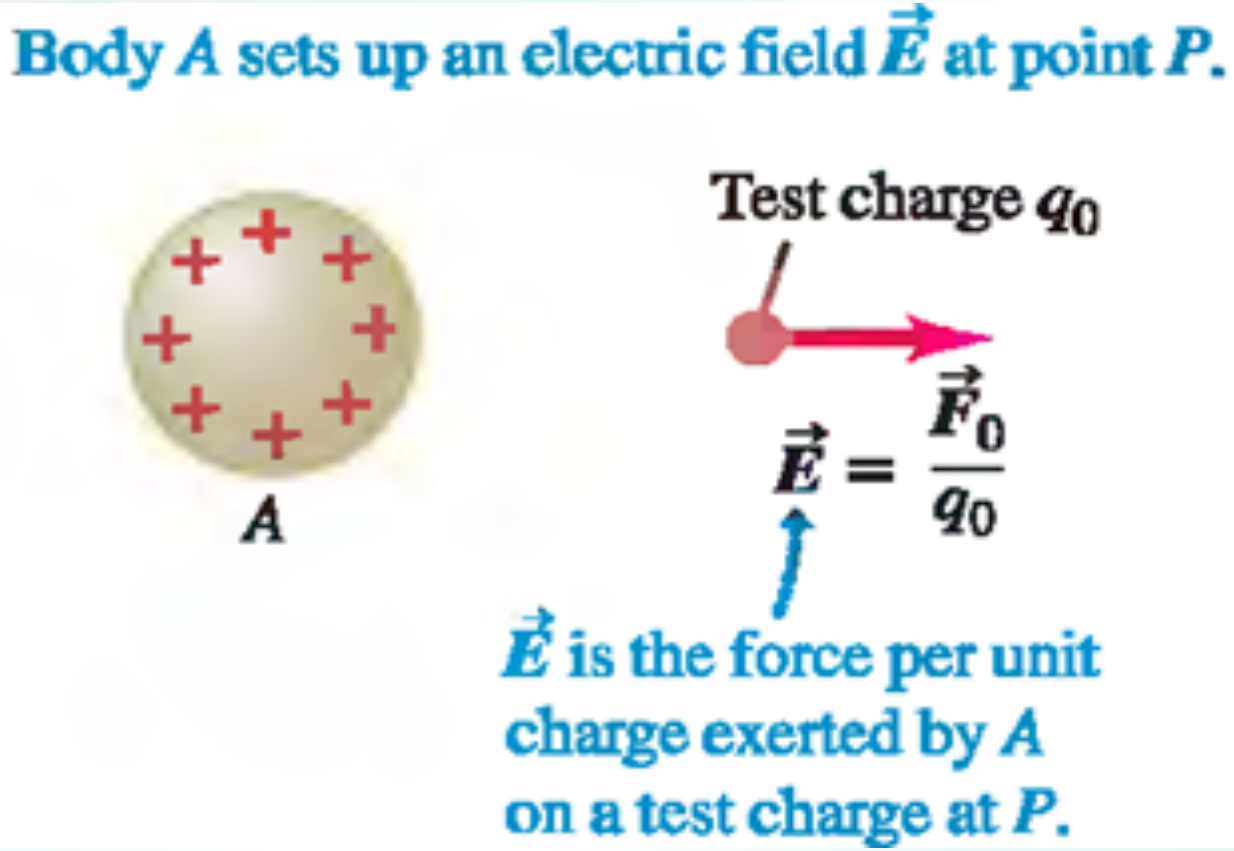
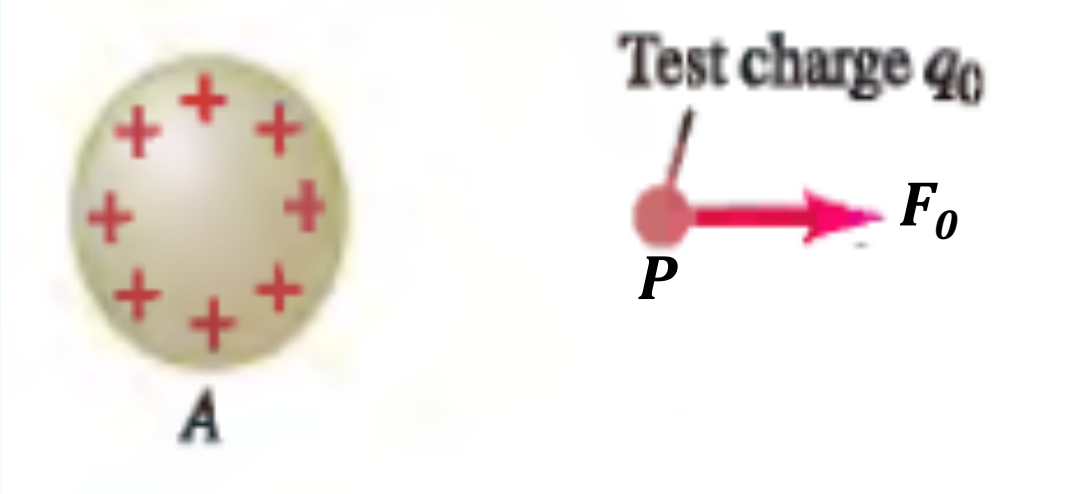


Figure 1. c) A charged body creates an electric field in the space around it. . From. *Sear's & Zemansky's University Physics With Modern Physics* (pp 699) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson.

Electric Field

- **It's actually the electric field that exerts the force on the charge.**
- **The charged particles DO NOT require the presence of another particle to produce an electric field. They produce the electric field by themselves.**



It is actually the electric field that exerts a force on the particles.

Figure 1. c) A charged body creates an electric field in the space around it. . From. *Sear's & Zemansky's University Physics With Modern Physics* (pp 699) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson

Electric Field

The particles are able to create an electric field as a result of the charge they carry. This somehow modifies the properties of the space around them.

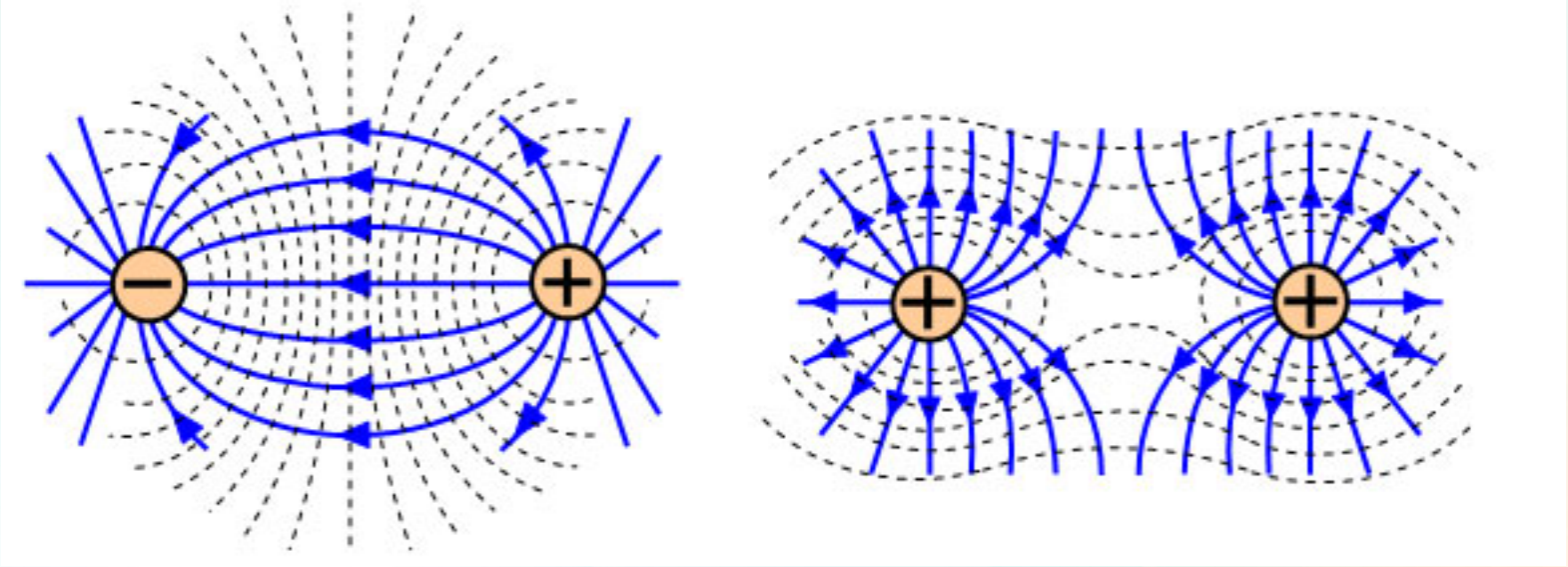


Figure 2. Electric Field Lines. Retrieved from https://brilliant.org/wiki/electric-field-lines/

Electric Field

Electric field is defined as the **force per unit charge** experienced by a **stationary point** charge in a **given location in a field.**

In equation:

$$\vec{E} = \frac{\vec{F}}{q}$$

where:

E – electric field in N/C

F – force in N

q – is the charge in C

Electric Field

The strength of an electric field E at any point may be defined as the electric force F exerted per unit positive electric charge q at that point, or simply E = F/q.

Using Coulomb's Law, the magnitude of the electric field at any given point around a charge can be determined.

$$E = \frac{F}{q} = \frac{\frac{kq_1q_2}{r^2}}{q} = \frac{kq}{r^2}$$

Where

E = electric field (N/C)
F = electric force (N)
q = charge (C)
r = distance between charges (m)

Sample Problem

1. The positive test charge in the figure shown has a magnitude of 3.0x10-8C and experiences a force of 6.0x10-8N.

Find the force per coulomb that the test charge experiences.

Predict the force that a charge of +12x10-8C would experience if it replaced the test charge.

Solution :

(a) $$\frac{F}{q_o} = \frac{6.0\times10^{-8}\,\mathrm{N}}{3.0\times10^{-8}\,\mathrm{C}} = 2.0\,\mathrm{N/C}$$

(b) $$F = (2.0\,\mathrm{N/C})(12.0\times10^{-8}\,\mathrm{C}) = 24\times10^{-8}\,\mathrm{N}$$

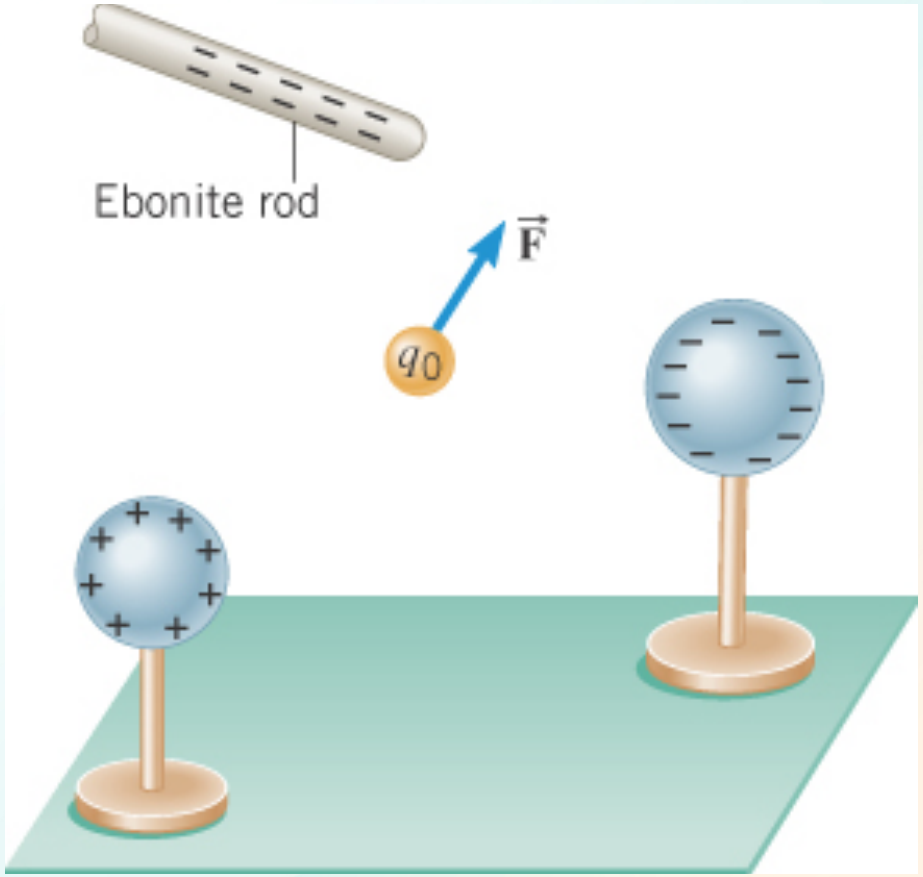


Figure 3. Electric Field. Retrieved from http://physicsskills.pbworks.com/f/E%2BForces%2Band%2BE%2BFields.pdf

DIRECTION AND STRENGTH OF ELECTRIC FIELD

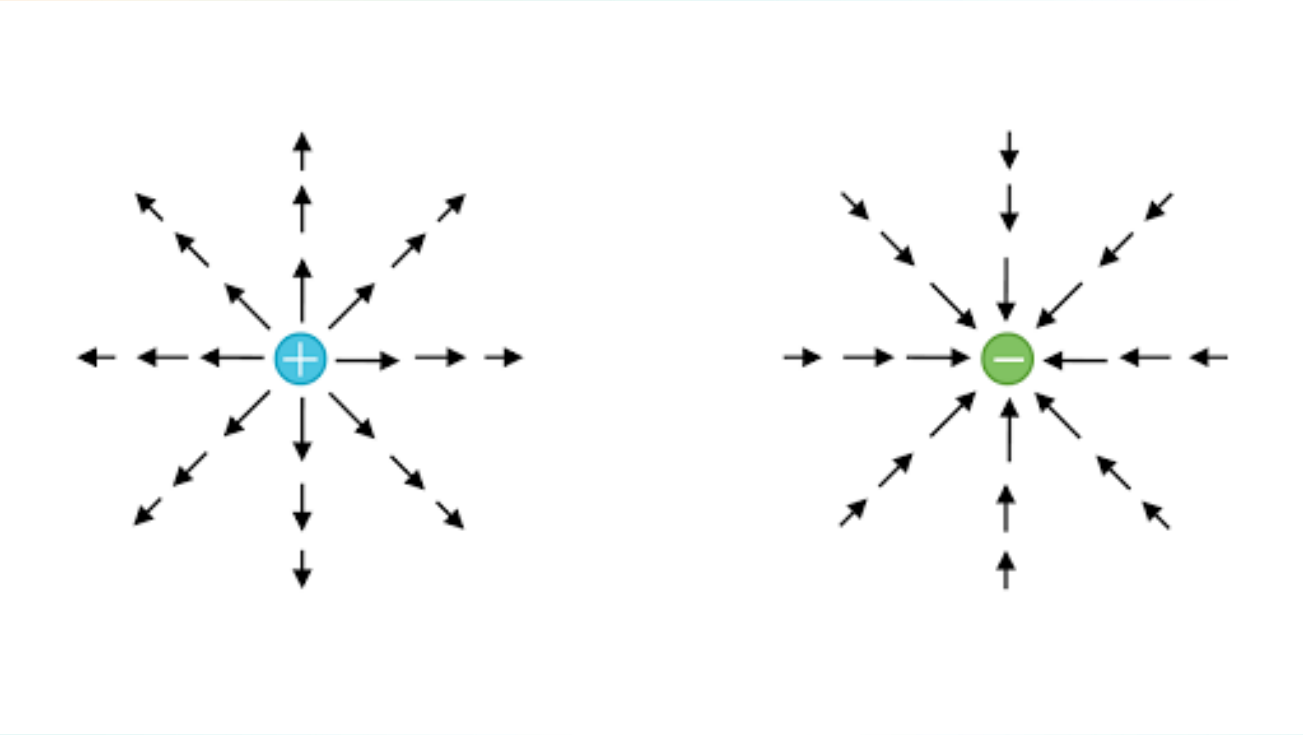


Figure 4. Equivalent representations of the electric field Retrieved from https://courses.lumenlearning.com/physics/chapter/18-5-electric-field-lines-multiple-charges/

The electric field **points away from a positive charge** and **toward a negative charge** and the electric field **strength/magnitude decreases with distance.**

Electric Field Lines

Michael Faraday (1791-1867) first introduced the concept of field lines. He called them "lines of force" but the term "field lines" is preferable.

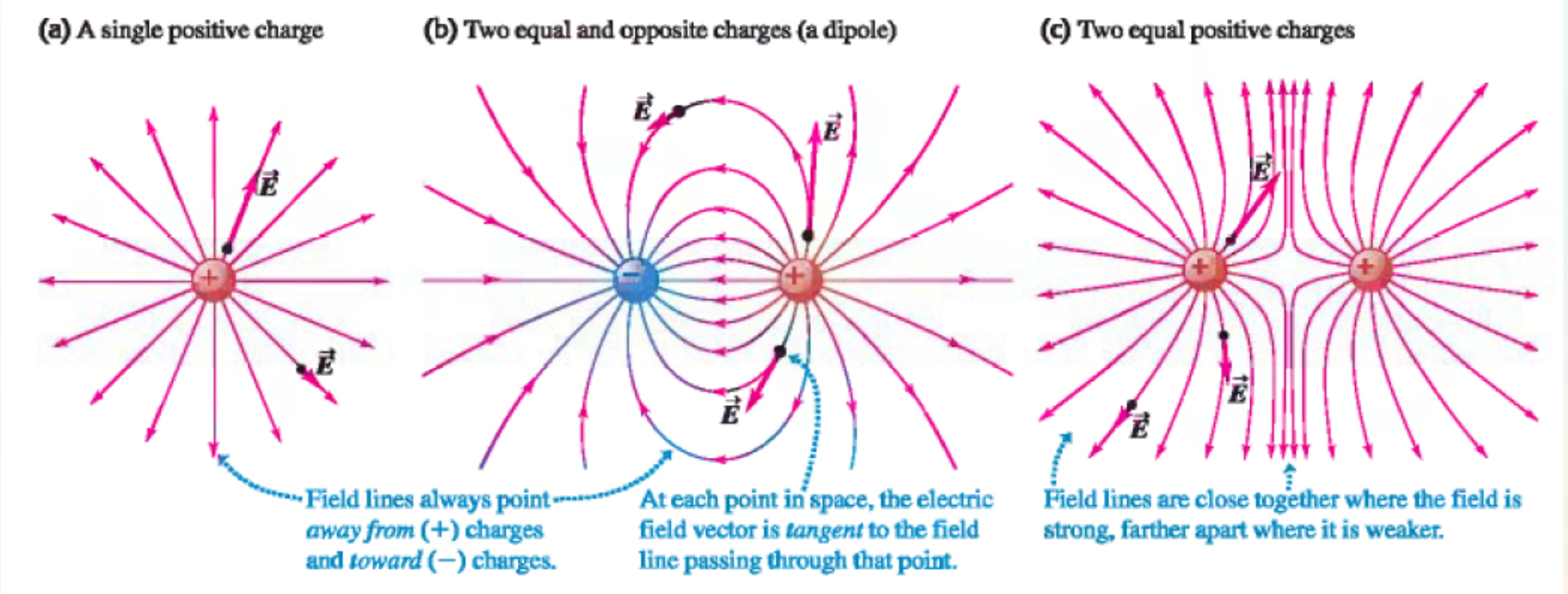


Figure 5. Electric field lines for three different charge distributions. From. *Sear's & Zemansky's University Physics With Modern Physics* (pp 699) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson.

Electric field lines for three different charge distributions. In general the magnitude of $\vec{E}$ is different at different points along a given field.

Electric Field Lines

An **electric field line** is an imaginary **line** or **curve** drawn through a region of empty space so that its tangent at any point is in the **direction of the electric field** vector at that point.

Electric Field Lines

The combination of 2 equal charges with opposite signs is called electric dipole.

The electric field lines are curved and are directed from the positive charge to the negative charge.

The direction of the electric field at any point is tangent to the field of line at any point (as shown by the arrow at arbitrary point P)

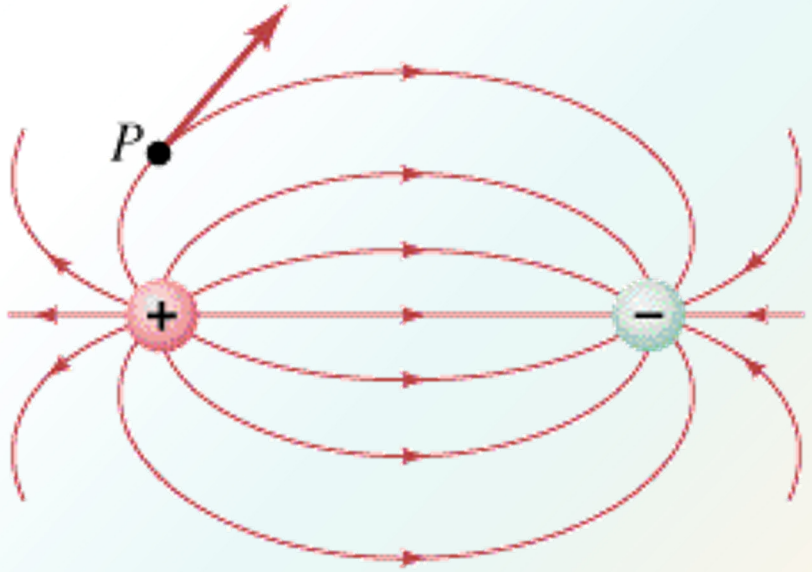
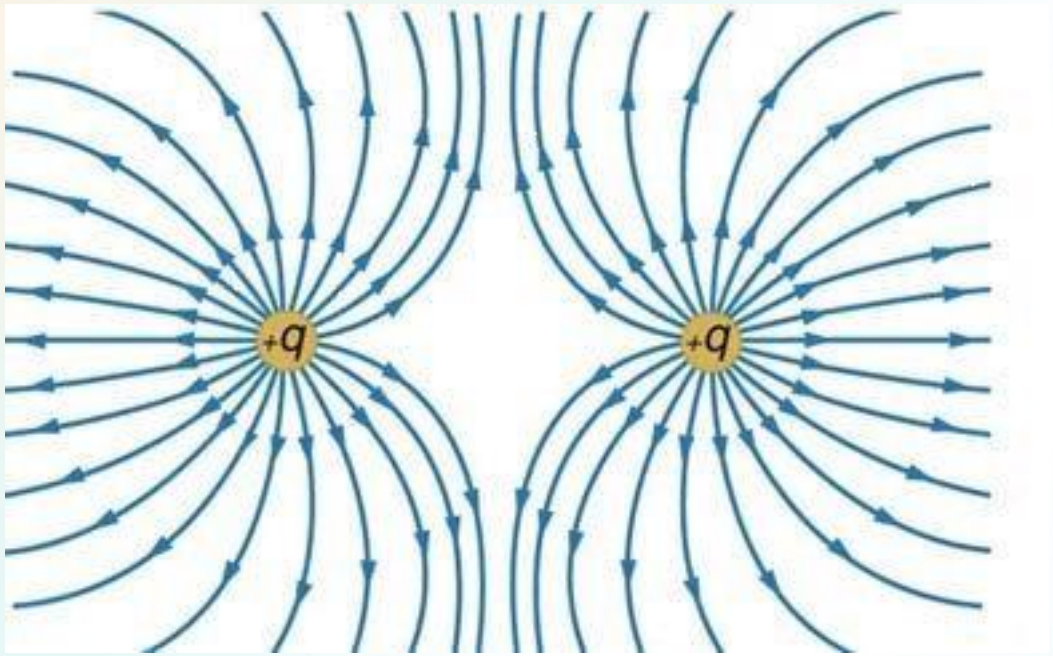


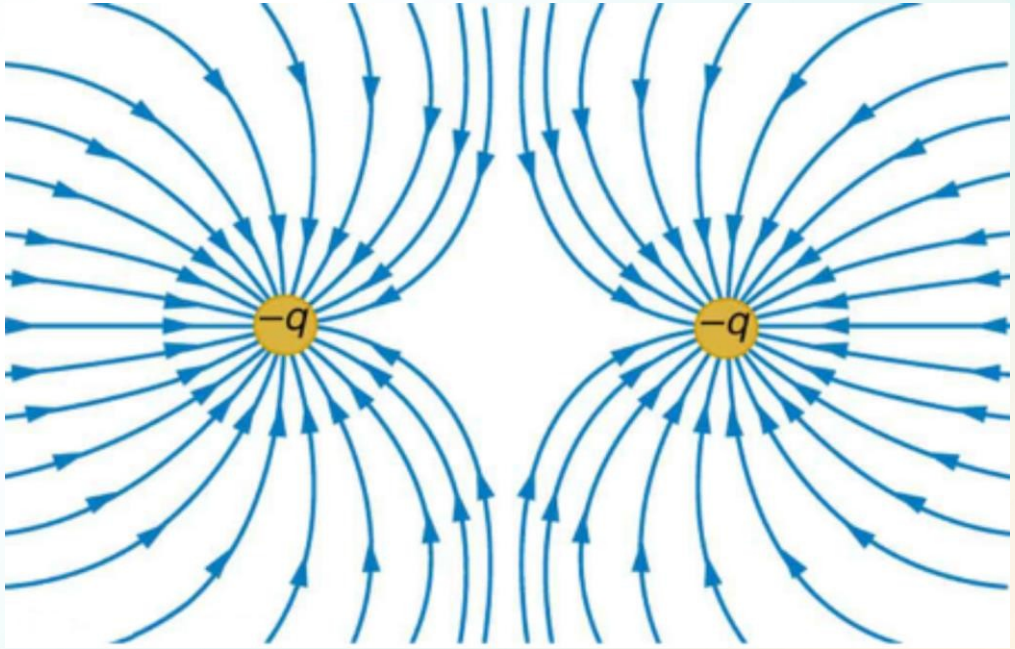
Figure 6. Electric Field Lines due to two equal charges of opposite signs.Retrieved from https://courses.lumenlearning.com/physics/chapter/18-5-electric-field-lines-multiple-charges/

The electric field lines due to two EQUAL charges of OPPOSITE signs, a combination known as **ELECTRIC DIPOLE.**

Electric Field Lines



Electric field lines for 2 EQUAL POSITIVE charges



Electric field lines for 2 EQUAL NEGATIVE charges

Figure 7 Electric Field Lines for 2 Equal Positive (Negative) charges. Retrieved from https://courses.lumenlearning.com/physics/chapter/18-5-electric-field-lines-multiple-charges/

Properties of Electric Field Lines

1. Electric field lines indicate the direction of the electric field; the field points in the direction tangent to the field line at any point

2. The lines are drawn so that the magnitude of the electric field E is proportional to the number of lines crossing unit area perpendicular to the lines. The closer together the lines, the stronger the field.

3. Electric field lines start on positive charges and end on negative charges; and the number starting or ending is proportional to the magnitude of the charge.

Note: Field lines do not cross because it would not make sense for the electric field to have 2 directions at the same point.

Electric Field Lines

The number of lines leaving a positive charge or entering a negative charge is proportional to the magnitude of the charge.

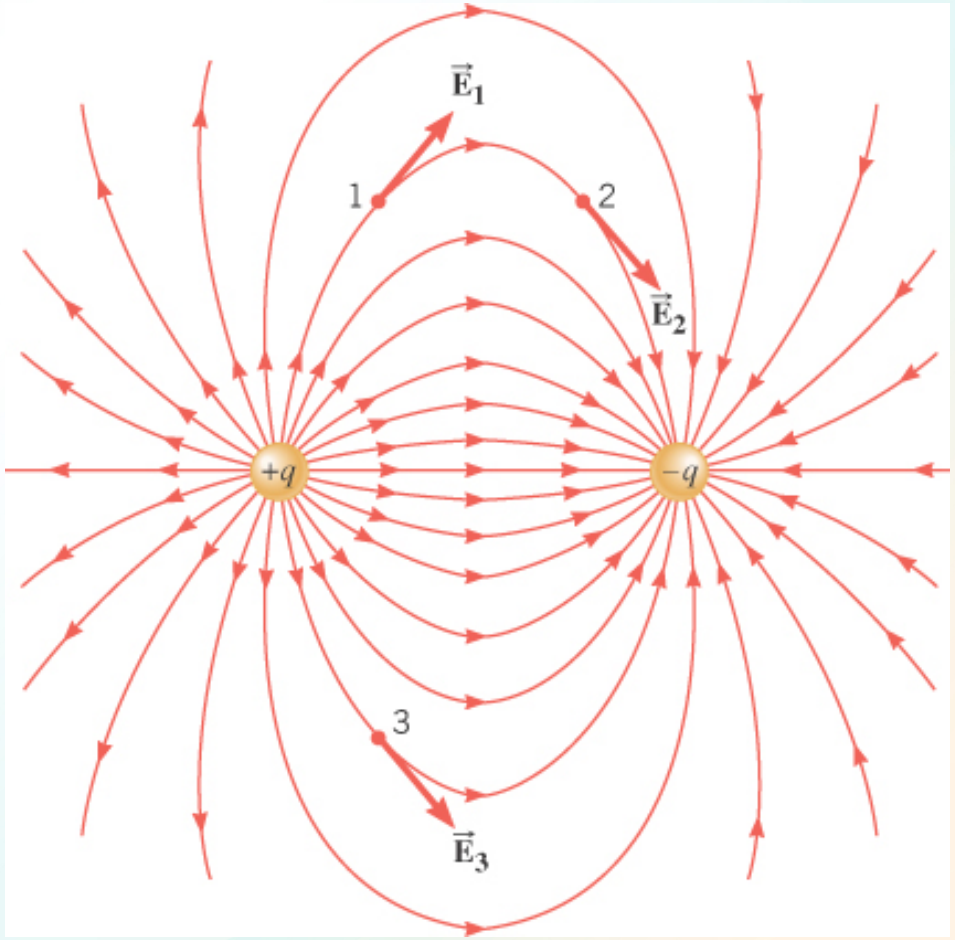


Figure 8. Electric Field Lines leaving a positive charge or entering a negative charge.. Retrieved from https://courses.lumenlearning.com/physics/chapter/18-5-electric-field-lines-multiple-charges/

Conceptual Sample

There are three things wrong with part (a) of the drawing. What are they?

Answer:

1) Field lines can never cross.
2) The number of electric field lines that end on the negative charges.
3) Spacing of electric field lines.

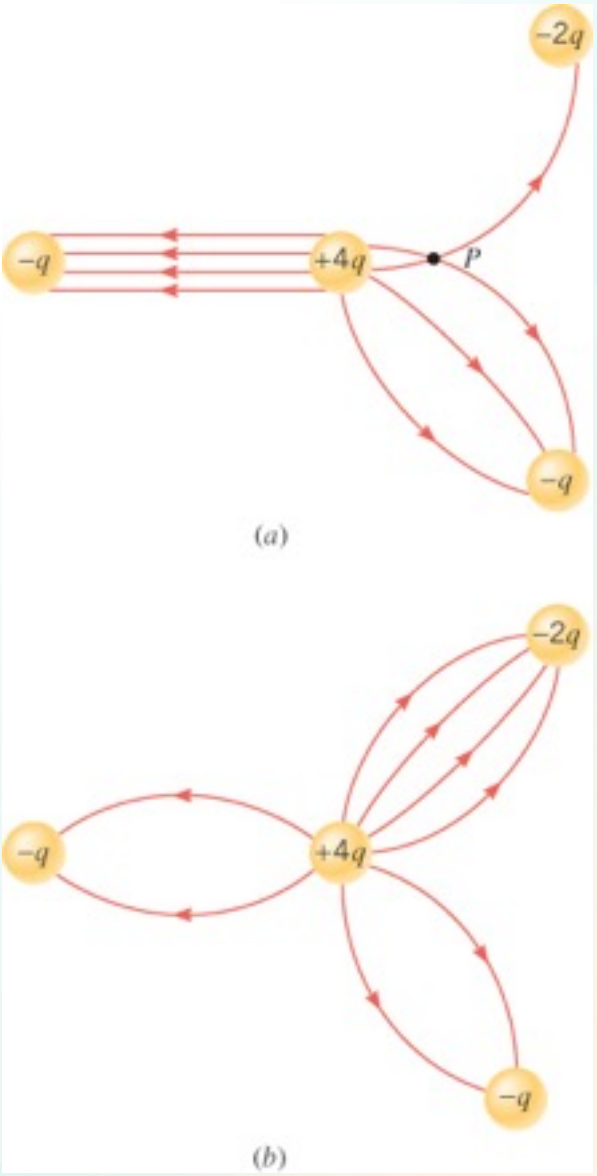


Figure 9. Conceptual Sample . Retrieved from https://courses.lumenlearning.com/physics/chapter/18-5-electric-field-lines-multiple-charges/

REFERENCES

Serway, R. A., & Jewett, J. W. (2019). Physics for Scientists and Engineers with Modern Physics Tenth Edition. Australia: Cengage Learning.

Young, H. D., & Freedman, R. A. (2016). Sear's & Zemansky's University Physics With Modern Physics. England: Pearson.

https://brilliant.org/wiki/electric-field-lines/

http://physicsskills.pbworks.com/f/E%2BForces%2Band%2BE%2BFields.pdf

https://courses.lumenlearning.com/physics/chapter/18-5-electric-field-lines-multiple-charges/

College Physics 2

Motion of Point Charges in Electric Field

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OBJECTIVES

- Describe the motion of charges in electric field
- Solve problems involving discrete and continuous charge distribution.

MOTION OF A CHARGE IN ELECTRIC FIELD

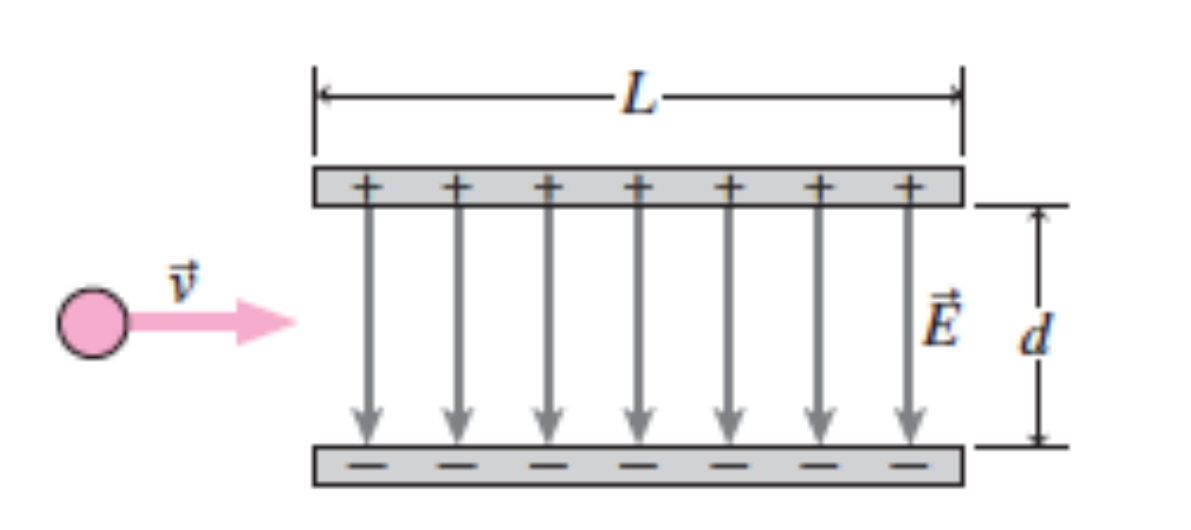


Figure 10. Electron moves in a Uniform Electric Field. From. *Essential University Physics* (pp 350) by Wolfson R.(2012). Addison-Wesley, Pearson

$$\boldsymbol{E = \frac{F}{q}}$$

$$F = qE$$

$$q = -e$$

$$F = -eE$$

$$F = ma$$

$$-eE = ma$$

$$\boldsymbol{a = -\frac{eE}{m}}$$

An electron moves in a uniform electric field

MOTION OF A CHARGE IN ELECTRIC FIELD

A positive charge moves in a uniform electric field

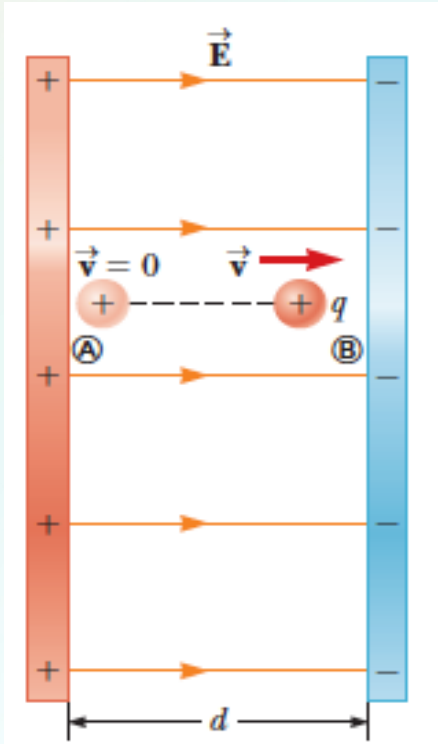


Figure 11. A positive point charge in a uniform electric field. From. *Physics for Scientists and Engineers with Modern Physics*(pp 711) by Serway, R. A., & Jewett, J. W. (2019). Australia: Cengage Learning.

$$\boldsymbol{E = \frac{F}{q}}$$

$$F = qE$$

$$F = qE$$

$$F = ma$$

$$qE = ma$$

$$\boldsymbol{a = \frac{qE}{m}}$$

ELECTRIC FIELD OF A POINT CHARGE

MAGNITUDE

If we place a small test charge q_0 at the field point P, at a distance r from the source point, the magnitude is given by Coulomb's Law:

$$\left[F = \frac{1}{4\pi\varepsilon_o}\frac{|qq_o|}{r^2}\right]\frac{1}{q_o}$$

$$E = \frac{1}{4\pi\varepsilon_o}\left(\frac{|q|}{r^2}\right) = k\frac{|q|}{r^2}$$

Where:

- E – Electric Field in (N/C)
- k – Coulomb constant
- k ≈ 9.0 x 10^9 Nm^2/C^2
- q – Charge in Coulomb (C)
- r – distance from the charge to point P in meters (m)
- F – Electrical Force in newton (N)
- q_0 – test charge in Coulomb (C)

ELECTRIC FIELD OF A POINT CHARGE

DIRECTION

(a) If q is positive, the force on the test charge is directed away from q.

(b) For the positive source charge, the electric field at P points radially outward from q.

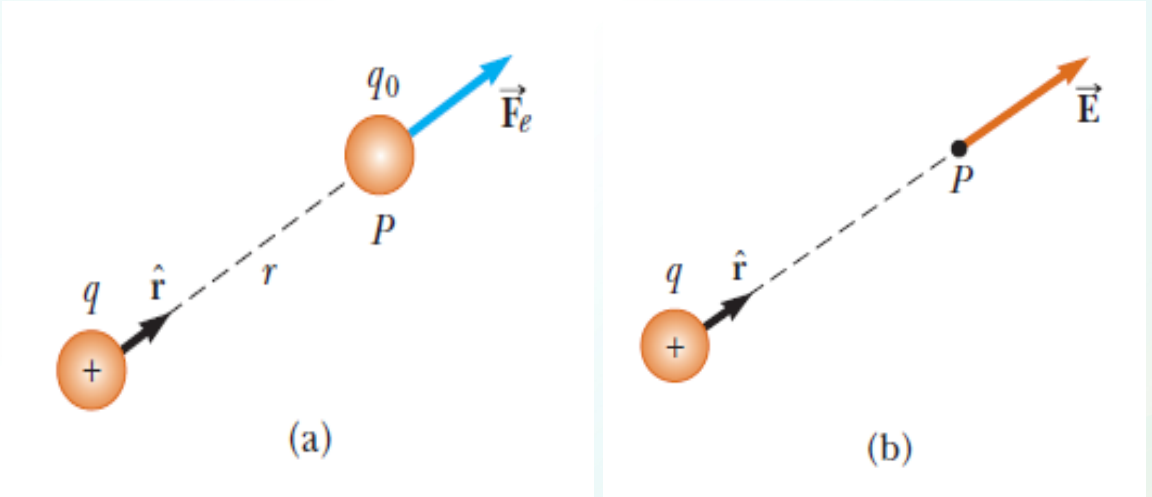


Figure 12. a) Force on a Test Charge. b) Electric Field of a Test Charge From. *Physics for Scientists and Engineers with Modern Physics*(pp 700) by Serway, R. A., & Jewett, J. W. (2019). Australia: Cengage Learning.

ELECTRIC FIELD OF A POINT CHARGE

DIRECTION

(c) If q is negative, the force on the test charge is directed towards q.

(d) For the negative source charge, the electric field at P points radially *inward* from q.

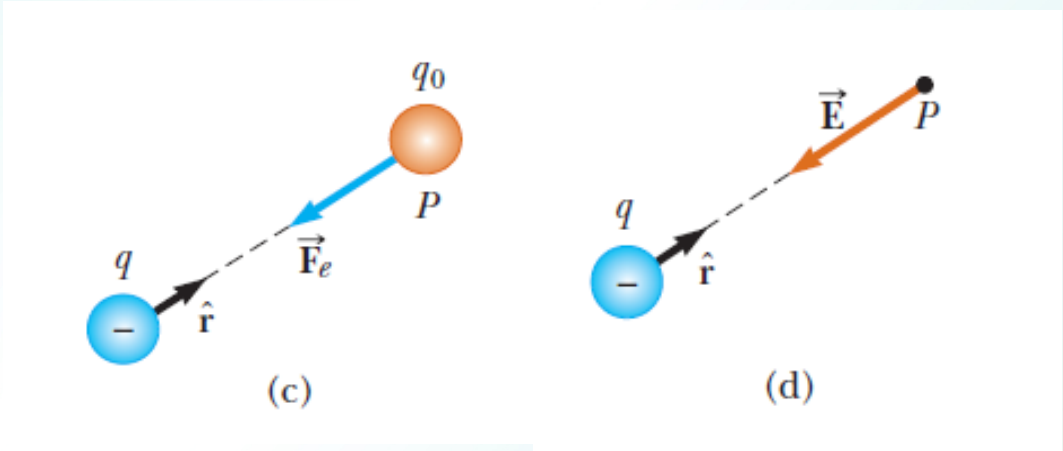


Figure 12. c) Force on a Test Charge. d) Electric Field of a Test Charge From. *Physics for Scientists and Engineers with Modern Physics*(pp 700) by Serway, R. A., & Jewett, J. W. (2019). Australia: Cengage Learning.

SAMPLE PROBLEMS

1. **What is the magnitude of the electric field at a point 2.0 m from a point charge q = 4.0 nC?**

SOLUTION

$$E = k\frac{|q|}{r^2}$$

$$= 9.0x10^9\ Nm^2/C^2\ \frac{|4x10^{-9}C|}{(2m)^2} = 9.0N/C$$

SAMPLE PROBLEMS

2. What is the magnitude of a point charge that would create an electric field of 5N/C at points 150 cm away?

SOLUTION

$$E = k\frac{|q|}{r^2}$$

$$\mathrm{q} = \frac{Er^2}{k}$$

$$\mathrm{q} = 1.25 \times 10^{-9}\ \mathrm{C}$$

ELECTRIC FIELD OF POINT CHARGES

If many charges are present in a given region, each charge contributes to the electric field at a known point. The Net electric field is the vector sum of all the fields of the charges that is,

$$\overrightarrow{E_T} = \overrightarrow{E_1} + \overrightarrow{E_2} + \cdots + \overrightarrow{E_n}$$

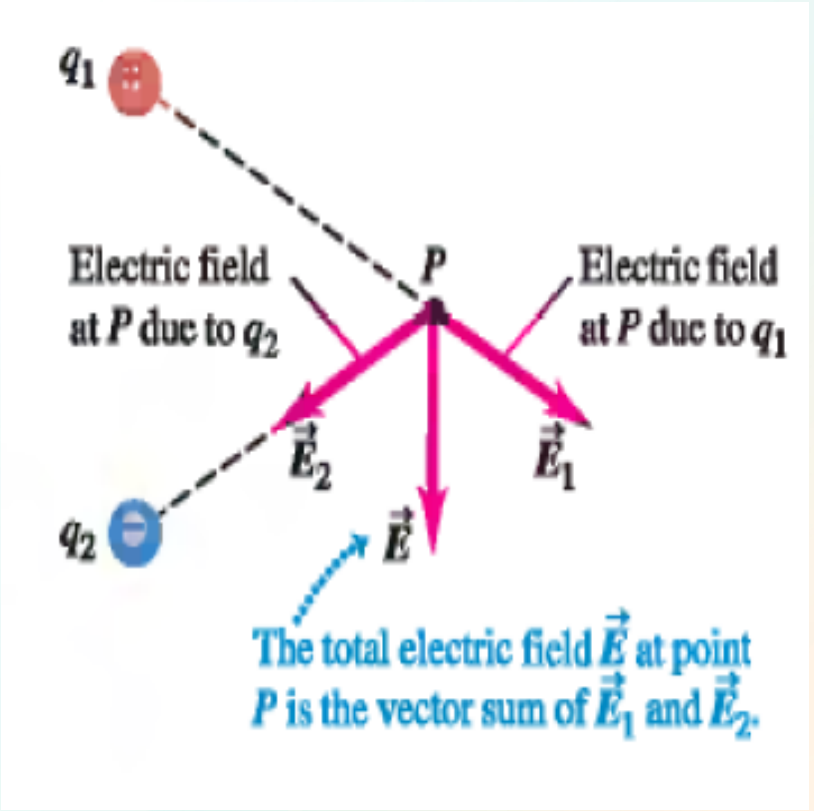
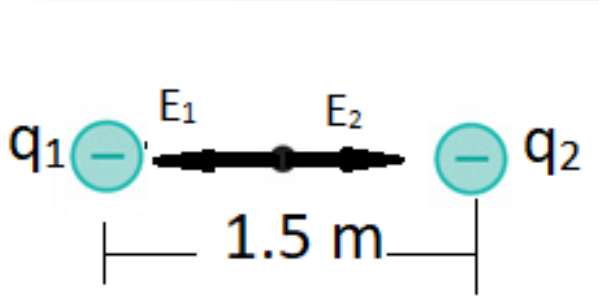


Figure 13. The total electric field. From. *https://www.miniphysics.com/wp-content/uploads/2010/11/electric-field-between-point-charges.png*

SAMPLE PROBLEMS

3. Two charges of -5.00 nC and -3.00 nC are 1.5 m apart. Compute for the electric field halfway between them.



SOLUTION

Let E_1 be the magnitude of electric field due to q_1 = -5.00nC

E_2 be the magnitude of electric field due to q_2=-3.00nC

$$E_1 = k\frac{|q_1|}{{r_1}^2}$$

$$E_1 = 9.0x10^9\ Nm^2/C^2\ \frac{|-5x10^{-9}C|}{0.75m^2} = 80N/C, \text{ left}$$

$$E_2 = k\frac{|q_2|}{{r_2}^2}$$

$$E_2 = 9.0x10^9\ Nm^2/C^2\ \frac{|-3x10^{-9}C|}{0.75m^2} = 48N/C, \text{ right}$$

$$E = -80\frac{N}{C} + 48\frac{N}{C} = -32\ N/C$$

REFERENCES

Serway, R. A., & Jewett, J. W. (2019). Physics for Scientists and Engineers with Modern Physics Tenth Edition. Australia: Cengage Learning.

Young, H. D., & Freedman, R. A. (2016). Sear's & Zemansky's University Physics With Modern Physics. England: Pearson.

https://www.miniphysics.com/wp content/uploads/2010/11/electric-field-between-point-charges.png